Zifei Dong

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Education

Vanderbilt University **Aug 2024 – May 2026**
Master of Data Science, GPA: 3.8/4 *Nashville, TN*

University of North Carolina at Chapel Hill **May 2021 - May 2024**
B.S. in Computer Science double major in Statistics and Analytics, Dean's List, GPA: 3.625/4 *Chapel Hill, NC*

Technical Skills

Programming: Python, SQL, Bash, R, Java, C++, MATLAB
LLM / NLP: PyTorch, Transformers, PEFT/QLoRA, SGLang, Instruction Tuning, Long-Context Modeling, Information Extraction, Text Classification, Evaluation Design
ML & Data Systems: TensorFlow, NumPy, Pandas, Linux, Multiprocessing, Spark, Hadoop, Weak Supervision, Data Augmentation, Robustness Evaluation

Selected Publication

- **Dong Z**, Wu W, Hao J, Chen T, Weng Z, Zhou B.
  AnyCXR: Human Anatomy Segmentation of Chest X-ray at Any Acquisition Position using Multi-stage Domain Randomized Synthetic Data with Imperfect Annotations and Conditional Joint Annotation Regularization Learning.
  Medical Image Analysis, under review. Available on arXiv.
  Contribution: Led project conception, end-to-end pipeline design, synthetic data generation strategy, model development, and experimental implementation.

Selected AIML Experience

Chinese Dialogue Evaluation and Companion Assistant Study [Showcase] **Dec 2025 – Present**
Independent Research Project *Nashville, TN*

- Designed a Chinese dialogue evaluation project based on **long-horizon** real-world conversations, focusing on **implicit user signals**, emotional support, clarification-seeking, and relational continuity.
- Distilled **16** representative cases with rubrics, organized into **4** user-state buckets covering confirmation-seeking, low-explicit-emotion expression, and calibration/challenge scenarios.
- Built a **semi-automated workflow** to transform private long-form conversations into patterns, error taxonomies, cases, rubrics, and strategy hypotheses for a public-facing evaluation asset.
- Conducted multi-model benchmarking across GPT, Claude, and Gemini, with an additional **multi-judge sensitivity analysis**; identified systematic differences in intent recognition, relational continuity, and calibration timing, and derived turn-level response strategy hypotheses for chatbot design.

Implicit Needs and Interaction Patterns in Human-AI Dialogue **Jan 2026 – Present**
Independent Research Project *Nashville, TN*

- Studied **recurring patterns** in implicit need recognition, response stance, search use, pacing, and closing strategy in **longitudinal** human-AI conversations, analyzing how differences in model style shape user trust and perceived task completion.

LLM-Based Financial Sentiment Extraction from SEC Filings **Jan 2026 – Present**
Research Collaborator, AllianceBernstein–Vanderbilt Joint Project *Nashville, TN*

- Contributed to a collaborative LLM pipeline for extracting factor-level signals from SEC filings and converting them into financially grounded sentiment predictions, primarily leading model training on **Qwen3-14B**.
- Trained and adapted LLMs for thematic factor extraction and factor-specific sentiment classification on **10-K**, **10-Q**, and **8-K** filings, using **Unsloth** and **QLoRA** for parameter-efficient fine-tuning and **SGLang** for scalable inference, evaluation, and experimentation.
- Extended the pipeline with **Chain-of-Thought (CoT)** supervision and **RLHF**-style alignment to improve reasoning consistency, sentiment calibration, and the downstream usefulness of model-generated financial signals.

AnyCXR Project, AIMP Lab, Northwestern University **April 2024 – Present**
Researcher, Remote *Evanston, IL*

- Led the design and development of **AnyCXR**, a robust anatomy segmentation framework for **chest X-rays from arbitrary viewpoints**, addressing severe acquisition and domain shift beyond standard clinical views.
- Built a **3D multi-stage synthetic data generation pipeline** over **8TB** of data from **50,188 CT scans** to simulate diverse projection geometries and improve cross-view generalization.

- Developed a **U-Net-based model** with **Conditional Joint Annotation Regularization** to train effectively on imperfect and partially labeled large-scale datasets.
- Achieved **94% mean Dice** across **54 anatomical structures** and improved downstream disease classification by **2%** through a segmentation-guided DenseNet pipeline.

Medical Image Denoising Research, AIMP Lab, Northwestern University **Dec 2025 – Present**

Researcher, Remote *Evanston, IL*

- Developed a **2.5D angiographic X-ray denoising and reconstruction** framework for low-dose imaging, targeting motion artifacts, structural distortion, and anatomically implausible outputs.
- Designed and compared multiple architectures, finding that a **VAE-based model without skip connections**, with **spatio-temporal attention** inserted after the encoder, produced the strongest denoising performance.
- Achieved over **36 PSNR** in denoising, with the VAE reconstruction backbone reaching over **31 PSNR**.

Ningbo Jinge Liangrui Asset Management Co. Ltd. **Jun 2025 – Aug 2025**

Quantitative Research Intern *Shanghai, China*

- Built and maintained an **end-to-end** pipeline for high-dimensional daily financial data spanning **10 years**, including data cleaning, alignment, feature generation, and model-ready dataset construction for deep learning–based signal research.
- Led computational optimization for data loading and model training by rewriting key **matrix operations** with **Numba** and **NumPy**, achieving a **10**× training speedup and enabling faster large-scale experimentation.
- Developed and evaluated a **Generalized Signature Neural Network** for next-year return prediction, achieving **1.2% R-squared** on the full dataset.

Additional Experience

Ultimate Consequences Lab, Vanderbilt University **March 2025 - Present**

Data Architect & Engineer, Part-Time *Nashville, TN*

- Architected and deployed an **end-to-end data pipeline** for a large-scale social science study, handling sensitive life-history and mortality data for **639 individuals** supporting an oral presentation.
- Engineered the core infrastructure, including designing a standardized **database schema**, developing automated **R crawlers (rvest/httr)** for five dynamic data sources, and building a modular workflow for **LLM-based** data enrichment and validation.
- Implemented software engineering best practices by refactoring legacy code in **R** to eliminate global variables and ensure reusability, enabling reliable collaboration among non-technical researchers and boosting overall **team efficiency by 50%**.

Dong Wei Lab, Tianjin University **August 2023 – April 2024**

Lab Research Assistant, Part-Time *Tianjin, China*

- Supported a publication on VR-based vocational skills learning by developing a **Unity3D** virtual training environment and implementing synchronized behavioral and physiological data collection with the **SteamVR SDK**.
- Processed survey, interview, and physiological data from **92 learners** using **Python** and **R**, and built an NLP pipeline with **NLTK** and **TextBlob** to extract sentiment and thematic signals for quantitative evaluation of VR-blended learning outcomes.

Liu Xinqiao Research Group, Tianjin University **August 2022 – May 2023**

Lab Research Assistant, Part-Time *Tianjin, China*

- Contributed to a peer-reviewed longitudinal study of **2,812 students** on stress and academic self-efficacy by designing and implementing cross-lagged statistical modeling workflows in **Python** and **R**.
- Led data preprocessing, statistical analysis, and visualization of longitudinal and model-pathway results, supporting interpretation and validation of the study's core findings.

Ningbo Jinge Liangrui Asset Management Co. Ltd. **May 2023 – August 2023**

Quantitative Research Intern *Shanghai, China*

- Built and evaluated time-series factor models for China A-Share (Python; linear models/ARIMA), supporting low-frequency factor screening, risk estimation, and back-testing.

North Carolina Pandemic Recovery Office **June 2022 - August 2022**

Software Engineer Intern *Raleigh, NC*

- Designed and automated a data processing pipeline using **pandas** and **shutil** to integrate OneDrive expense data, reducing the audit team's weekly workload by **1 day**.